

Product Requirements (Updated to Match Current Project)

This requirements list reflects the implemented Product MVP in this repository. It scopes the project to the current Next.js internal tool, employee authentication, seeded Target product data, heuristic recommendations, protected app pages, and demo-ready dashboards.

ID	Requirement	Description	Team Member
FR-1	Employee login	The system shall allow users to log in with an employee ID and password through the `/login` page.	Hadent
FR-2	Demo account creation	The system shall provide a hidden `/internal-signup` route for creating class-project demo employee accounts.	Hadent
FR-3	Protected internal routes	The system shall prevent unauthenticated users from accessing protected internal pages under the app layout.	Hadent
FR-4	Authenticated app shell	The system shall display a branded internal app shell with a sidebar, header, employee identity, and logout support after login.	Hadent
FR-5	Dashboard overview	The system shall provide a `/dashboard` page summarizing the PA4 MVP services and linking to evidence routes.	Hadent
FR-6	Navigation tabs	The system shall provide sidebar tabs for Dashboard, Store Performance, Inventory, Competitor Analysis, Product Analysis, Notifications, Messages, Store Layout, and Settings.	Hadent
FR-7	Web scraper seed process	The system shall include a repeatable seed script that imports Target sample product data into the database.	Hadent
FR-8	Seed process reporting	The seed script shall report processed, inserted, updated, skipped, and elapsed counts when product data is loaded.	Hadent
FR-9	Database schema	The system shall define Drizzle-managed database tables for authentication, sessions, direct messages, Target products, and recommendation runs.	Hadent
FR-10	Database migrations	The system shall provide committed Drizzle migrations that teammates can apply with the project migration command.	Hadent
FR-11	Product picker	The Product Analysis page shall let users select a seeded Target product from a searchable product selector.	Simra
FR-12	Product details	The Product Analysis page shall display the selected product title, category, image, rating, review count, seller, price, breadcrumbs, and description when available.	Simra
FR-13	Product data completeness	The Product Analysis page shall calculate and display a completeness percentage based on populated seeded product fields.	Simra
FR-14	Product discount signal	The Product Analysis page shall calculate and display a discount signal when initial and final prices show a markdown.	Simra
FR-15	Currency handling	The Product Analysis page shall normalize seeded currency values such as `\$` to a valid display currency before formatting prices.	Simra
FR-16	Recommendation generation	The system shall generate related product recommendations for a selected product using the local heuristic recommendation service.	Simra
FR-17	Recommendation scoring	The recommendation service shall rank products using category overlap, price proximity, rating quality, and dataset recommendation hints.	Simra
FR-18	Recommendation reasons	The Product Analysis page shall display explainable reasons for each recommended product match.	Simra
FR-19	Recommendation persistence	The system shall store recommendation run results in the database for traceability.	Simra
FR-20	Recommendation API	The system shall expose recommendation results through `GET /api/recommendations/[productId]`.	Simra
FR-21	Product Analysis fallback	The Product Analysis page shall show a setup state if seeded product data or recommendation data is unavailable.	Sangeetha
FR-22	Invalid product fallback	The Product Analysis page shall recover from an invalid `productId` query by falling back to the first seeded product when possible.	Sangeetha

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FR-23	Store Performance page	The system shall provide a `/store-performance` tab showing store sales, customer count, refund/loss metrics, sales performance chart, and top products.	Sangeetha
FR-24	Inventory page	The system shall provide an Inventory page showing product inventory, on-hand units, incoming stock, reorder need, price, reviews, sales history, and return rate using demo UI data.	Sangeetha
FR-25	Competitor Analysis page	The system shall provide a Competitor Analysis page comparing Target and competitor pricing, stock, sales, sentiment, and AI-style decision support readouts.	Sangeetha
FR-26	Competitor visualization	The Competitor Analysis page shall display visual comparisons using charts, scorecards, and competitor summary cards.	Sangeetha
FR-27	Store Layout page	The system shall provide a Store Layout page for comparing current and recommended floor plan concepts.	Sangeetha
FR-28	Notifications page	The system shall provide a Notifications page for product, restock, pricing, and social-style alert messaging.	Sangeetha
FR-29	Messages page	The system shall provide a Messages page for reviewing faculty or internal conversations through the app shell.	Sangeetha
FR-30	Settings page	The system shall provide a Settings page for configuring organization profile, coverage, AI behavior, notifications, privacy, and interface preferences.	Sangeetha
FR-31	Direct message API	The system shall include API routes for reading messages and message users.	Hasnain
FR-32	App route redirect	The root route shall redirect authenticated users to `/dashboard` and unauthenticated users to `/login`.	Hasnain
FR-33	Session enforcement	The protected app layout shall require a valid session before rendering internal pages.	Hasnain
FR-34	Demo-friendly placeholders	Pages without full backend integrations shall clearly present demo UI or placeholder content instead of claiming live enterprise integrations.	Hasnain
FR-35	Build verification	The project shall support `npm run lint` and `npm run build` as verification commands for the implemented MVP.	Hasnain
FR-36	Local development	The project shall run locally through the Next.js development server using the configured environment variables.	Hasnain
FR-37	Environment configuration	The system shall read database and authentication configuration from local environment files such as `.env` or `.env.local`.	Hasnain
FR-38	MVP service evidence	The project shall document the Web Scraper, Database, Authentication, and AI Recommendation service mappings in project documentation.	Hasnain
FR-39	Implementation documentation	The project shall include documentation describing the actual implementation approach, technology stack, and verification steps.	Hasnain
FR-40	Deployment documentation	The project shall include documentation describing the current demo deployment process, setup steps, route access, and known blockers.	Hasnain